THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 8 — SCIENCE FULL SYLLABUS PAPER SET 08

Full Syllabus (Chapters 1–13) | RBSE Pattern | Session 2026–27

Time: 3 Hours 15 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 80**

General Instructions:

1. All questions are compulsory.
2. Q. 1–20 are Objective Type (1 mark each).
3. Q. 21–28 are Very Short Answer Type (2 marks each).
4. Q. 29–36 are Short Answer Type (3 marks each).
5. Q. 37–40 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 35 to 40.
7. Write the question number before each answer.
8. Draw neat, labelled diagrams wherever required.

SECTION — A (Objective Type Questions) 20 × 1 = 20

Choose the correct option and write it in your answer book.

1. The root image on left-hand pages symbolises: 1

(a) a deep foundation of knowledge and heritage
(b) a fruit
(c) a game
(d) fast travel

2. Repeating an experiment several times makes results more: 1

(a) random **(b)** reliable **(c)** hidden **(d)** colourful

3. Which is a unicellular organism? 1

(a) Human **(b)** Amoeba
(c) Mango tree **(d)** Mushroom cap only

4. Dengue is prevented by: 1

(a) wearing wet clothes
(b) sharing towels
(c) eating fried food
(d) avoiding mosquito breeding in still water

5. The first vaccine was developed by: 1

(a) Louis Pasteur **(b)** Edward Jenner
(c) Robert Hooke **(d)** Alexander Fleming

6. The heating element of an electric heater is made of: 1

(a) nichrome **(b)** plastic **(c)** copper **(d)** iron nail

7. Rechargeable batteries are used in: 1

(a) mobile phones and laptops **(b)** compasses
(c) torches only **(d)** wall clocks only

8. Muscular force helps in: 1

(a) only walking **(b)** only lifting
(c) only jumping **(d)** chewing food and pumping blood

9. A rubber sucker sticks to a wall because: 1

(a) magnetism **(b)** outside air pressure is more than inside
(c) friction only **(d)** glue is present

10. Thunder is the loud sound produced when: 1

(a) lightning heats and expands air **(b)** wind stops
(c) clouds collide like solids **(d)** rain falls

11. The temperature at which a solid melts at atmospheric pressure is its: 1

(a) freezing point of vapour **(b)** melting point
(c) sublimation limit **(d)** boiling point

12. Which set has only compounds? 1

(a) Air, sea water, brass **(b)** Water, carbon dioxide, sodium chloride
(c) Water, oxygen, salt **(d)** Iron, sulfur, gold

13. Brass is an alloy of: 1

(a) iron and carbon **(b)** copper and tin **(c)** copper and zinc **(d)** gold and silver

14. Volume of a liquid is measured with a: 1

(a) balance **(b)** measuring cylinder
(c) spring balance **(d)** clock

15. A ray of light makes an angle of 65° with the normal to a plane mirror. The angle of reflection is: 1

(a) 25° **(b)** 65° **(c)** 115° **(d)** 130°

16. Earth's rotation defines a: 1

(a) month **(b)** season **(c)** day **(d)** year

17. Non-living components of an ecosystem are: 1

(a) consumers **(b)** abiotic components
(c) producers **(d)** biotic components

18. Which is a major threat to ecosystems? 1

(a) planting trees **(b)** afforestation **(c)** deforestation **(d)** composting

19. In plants, the zygote develops into a: 1

(a) pollen **(b)** leaf **(c)** root hair **(d)** seed

20. Atmosphere, hydrosphere, geosphere and biosphere: 1

(a) act separately **(b)** do not affect each other
(c) are all gases **(d)** interact and sustain life

SECTION — B (Very Short Answer Type) 8 × 2 = 16

Answer in about 10–15 words each.

21. What is the function of the cell wall? 2

22. Give two measures to prevent dengue. 2

23. Why should used batteries be recycled? 2

24. Why are roofs blown away during high-speed winds? 2

25. Write the word equation for heating iron and sulfur. 2

26. State how density changes with temperature. 2

27. Define parasitism with an example. 2

28. What is a zygote? 2

SECTION — C (Short Answer Type) — 8 × 3 = 24

Answer in about 30–40 words each.

29. Suggest two questions about the Moon that can be investigated by observing the sky. 3

30. How is friction useful? Give three examples and say what would happen without it. 3

31. What happens to the interparticle space when a gas is cooled? Explain. 3

32. A light ray incident at 50° falls on a mirror. Find the angle between incident and reflected rays. 3

33. How do satellites help in weather forecasting and disaster management? 3

34. Differentiate between a plant greenhouse and the greenhouse effect on Earth. 3

35. Describe the activity that shows a current-carrying wire deflects a compass needle. 3

OR

Why are electromagnets used in cranes instead of permanent magnets? 3

36. Draw a food chain with four trophic levels and label them. 3

OR

Describe the pyramid of numbers using millet, mouse, snake and eagle. 3

SECTION — D (Long Answer Type) — 4 × 5 = 20

Answer in about 80–100 words each.

37. Discuss the symptoms, causes and prevention of tuberculosis and hepatitis A. 5

OR

Describe the role of hygiene, clean water and sanitation in preventing communicable diseases with three examples. 5

38. Describe the formation of a cyclone and its structure. 5

OR

Explain why a rubber sucker sticks and why it becomes hard to remove, using the idea of pressure differences. 5

39. A cold fizzy drink is opened in a hot place. Explain what happens with solubility of gases. 5

OR

Write a note on the importance of solutions in nature and industry. 5

40. Explain how human activities affect ecosystems and what individuals can do. 5

OR

Compare terrestrial and aquatic ecosystems with examples, abiotic factors and organisms. 5

END OF QUESTION PAPER — Total: 80 Marks

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